Prestige vs. Price: Spatial Capitalization of Higher Education at State Borders

Evidence from a Difference-in-Discontinuities Design (2012–2024)

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April 10, 2026

Roadmap

- **Setup & Hypotheses:** Spatial boundaries of public university tuition and caliber at state borders.
- **Empirical Strategy:** The Spatial Difference-in-Discontinuities (DiDC) method.
- **Data & Curation:** NBER TAXSIM microsimulations and MSE-optimal bandwidth selection.
- **Results:**
  - Baseline
  - Heterogeneity
  - Robustness
- **Future steps**
- **Conclusion**

Background: Higher Education at the State Border

- State borders create sharp, discontinuous "cliffs" in university access.
- Crossing a state line changes one's access to:
  1. **Financial Subsidies:** Tens of thousands of dollars in in-state tuition discounts.
  2. **Prestige:** Potential access to elite, nationally-renowned public flagship institutions. In-state applicants generally have higher acceptance rates.
- **The Question:** Do households spatially sort to capture these benefits, and is this access capitalized into local housing markets?

Hypotheses

Hypothesis 1: Financial Cost Driven

Housing markets price the pure financial savings of the college "tuition cliff." Families sort to capture cheaper in-state pricing.

Hypothesis 2: Prestige Option Value

Housing markets price the *option value* of access to elite public flagship tiers (e.g., Top 50 institutions), regardless of the exact financial cost.

Theoretical Framework: Spatial Supply and Demand

- **Setup:** Two states (A and B) separated by a border, a local market is formed at the border.
- **Mechanism:** Housing supply $S(P)$ is fixed in the short run at exactly the border line.
- Residential demand $D(P, A_s)$ depends on housing price P and state-specific amenities $A_s = f(\text{Tuition}_s, \text{Caliber}_s)$.
- **Intuition:** If State A randomly discounts its in-state tuition or gains a Top 50 university, its amenity value A_A spikes.
- Demand shifts across the border from State B into State A. Since supply at the border is inelastic, land prices in A must rise discontinuously relative to B to clear the market.

Literature & Contribution

- **Tiebout Sorting (1956) & School Capitalization:** Black (1999) proves housing markets capitalize local K-12 school quality at attendance boundaries.
- **The Gap:** We know little about whether families sort at the *state level* for higher education.
- **Contribution:**
  - Extends Tiebout sorting from K-12 to the higher-education market.
  - Overcomes policy endogeneity typical of macro-state comparisons by using a high-resolution Spatial Difference-in-Discontinuities (DiDC) estimator across 448k estimation observations.

Empirical Strategy: Spatial DiDC

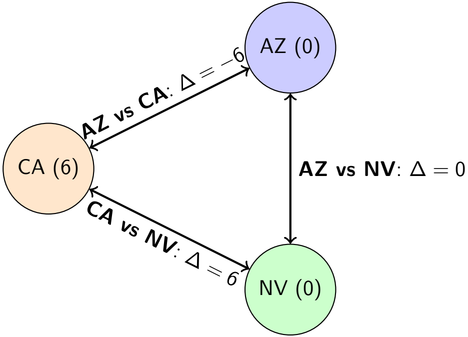
Econometric Specification

$$\ln(P_{zt}) = \beta(S_i \times \Delta\text{Edu}_{st}) + \gamma X_{zt} + \alpha_{sy} + \epsilon_{zt}$$

- S_i: Spatial dummy (1 if Treated Side, 0 if Control Side).
- ΔEdu_{st}: The dyadic gap in Tuition or Caliber.
- γX_{zt}: Time-varying controls (Federal & state income tax, property taxes, local GDP, unemployment).
- α_{sy}: **Border-Segment-by-Year Fixed Effects**. Absorbs all cross-border temporal and unobserved spatial shocks specific to that exact junction.
- **Two-Way Clustered Standard Errors** (ZCTA and Border Dyad).

Spatial Dummy Assignment

The Assignment Problem: For any state dyad, which state is $S_i = 1$? We arbitrarily sort alphabetically. The parameter math flawlessly absorbs the polarity.



- If CA is control ($S_{AZ} = 1$), $\Delta = -6$. If CA is treated ($S_{CA} = 1$), $\Delta = +6$.
- Switching the dummy assignment does not affect the β coefficient.

Data Sources & Variable Construction

Table: Summary of Final Estimation Panel (2012–2024)

Variable Category	Data Source	Specific Variable Construction
Housing Costs	**Redfin**	Median Price Per SqFt (ZCTA level)
Tuition Cliffs	**IPEDS**	In-State Tuition Gaps (Δ) between State Flagships
Institution Caliber	**U.S. News**	Historical Ranking Thresholds (Top 20, 50, 100)
Income Tax Shield	**NBER TAXSIM**	Simulated Federal & State Liability ($100k HH)
Macro Controls	**BEA, BLS**	County/ZCTA Real GDP and Unemployment
Local Policy	**ACS / TAXSIM**	Effective Property Tax Rates

Data Curation: Housing & Education Policy

1. **Housing Outcomes (2012–2024)**
   - ZCTA-level panel from Redfin (Median Price per Square Foot).
   - Derived distances to the boundary (d_{ic}) for precise spatial sorting.
2. **Higher Education Policy**
   - **Financial:** IPEDS panel for weighted state tuition averages.
   - **Prestige:** U.S. News & World Report historical rankings, algorithmically binned into Top 20, Top 50, and Top 100 state flagship caliber indicators.

Data Curation: NBER TAXSIM (The Tiebout Controls)

- **The Problem:** High-amenity states charge high taxes. Failing to control for tax shifts across the border thoroughly poisons the tuition coefficient.
- **The Solution (NBER TAXSIM):**
  - We simulated a standardized "Median Household" ($100k income, married, 2 dependents).
  - Fed this identical family through the tax code of all 50 states natively for every year.
  - Isolated *pure statutory tax liability variation* without endogeneity from actual local wage or employment differences.

Methodology: Border Segmentation & Bandwidth

- **The Problem:** Comparing across entire state borders confounds climate and economy.
- **Segmentation:** We fracture the US state-border network into discrete, evenly distributed segments ($\approx$ 50km bands).
- **Bandwidth (h^*):** We compute data-driven, MSE-optimal geographic target bandwidths locally using Triangular Kernels (via `rdrobust`).
  - **Efficiency:** Filters 2.7M raw observations down to a 448k estimation sample (16% retention).
  - **Mean Bandwidth:** $h^* \approx 63.33$ km.
  - **Failsafe:** Global fallback of 50km applied to sparse rural segments to ensure spatial completeness.
- This forces comparisons strictly between immediate cross-state neighbors operating in identical macroeconomic geographies.

Summary Statistics (Estimation DiDC Panel)

Table: ZCTA-Level Estimation Sample Properties (Filtered to Bandwidth)

Variable	**Observations**	**Mean**	**Std. Dev.**
Median Price Per SqFt ($)	268,224	163.22	367.17
Distance to Border d_{ic} (km)	448,941	0.43	36.82
Δ In-State Tuition Gap ($)	448,941	-163.74	3390.95
Δ Top 50 Caliber Gap (Count)	344,411	-0.11	0.88
Effective Property Tax Rate	407,324	0.012	0.007
Unemployment Rate (%)	444,795	5.071	2.174
Real GDP Per Capita ($)	444,795	58,720	1,050,974
TAXSIM Federal Income Tax ($)	419,176	198.88	6754.56
TAXSIM State Income Tax ($)	419,176	1959.85	1924.20

- Taxes simulated for a $100k median household; GDP and Unemployment captured at ZCTA-level.

Result I: The "Tuition Illusion"

Table: Capitalization of Tuition (Dependent Var: log PPSF)

	(1) Spatial RD	**(2) Full DiDC**
Tuition Gap LATE (\$1k)	-0.043^{*}	-0.008
	(0.022)	(0.013)
Effective Property Tax Rate		-19.494^{***}
		(3.081)
Income Tax / Macro Covariates		✓
Fixed Effects	None	Seg_ID × Year
Standard Errors	Single C.	Two-Way C.
Observations	196,632	128,420

- **Finding:** Raw Spatial RD assumes cheaper tuition drives higher prices (Col 1). Saturated DiDC models prove this is an illusion driven entirely by omitted state tax structures (Col 2).

Result II: The Joint Capitalization Model

Table: Joint Model: Tuition vs. Top 50 Prestige (Dep Var: log PPSF)

	(1) Base DiDC	**(2) Full DiDC (With Taxes)**
Caliber Gap LATE (Top 50)	0.093^{**}	$\mathbf{0.098}^{***}$
	(0.041)	(0.031)
Tuition Gap LATE ($1k)	-0.026^{***}	-0.008
	(0.009)	(0.013)
TAXSIM Covariates		✓
Fixed Effects	Seg $\times$ Year	Seg $\times$ Year
Standard Errors	Two-Way C.	Two-Way C.
Observations	135,402	128,420

- **Concrete Magnitude:** Gaining in-state access to a Top 50 university increases local housing prices by a precisely estimated **9.8%** ($p < 0.01$) exactly at the state border.

Exploring Option-Value Non-Linearities

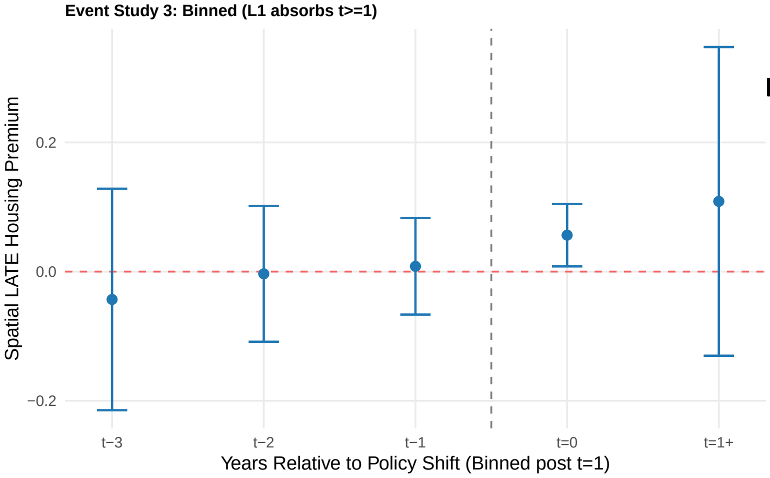
- Does the prestige tier matter? Yes. We observe a structural inverse U-shape in the willingness to pay for elite access.

Table: LATE Premium by Flagship Caliber Tier (Full DiDC / Two-Way SE)

Caliber Tier	**Coefficient**	**Std. Error**
Top 100 Gap (Generic Publics)	0.028	(0.027)
Top 50 Gap (Elite Flagships)	**0.098*****	(0.031)
Top 20 Gap (Hyper-Elite)	0.010	(0.048)

- **Intuition:** Top 100 is too broad to induce cross-state sorting. Top 20 is mathematically too sparse geographically to generate signal. Top 50 bounds the precise threshold of the market's psychological valuation.

Event Study: Validation & Market Efficiency



Key Takeaways:

- **Zero Pre-Trends:** Leads ($t < 0$) firmly cross zero. Prevents reverse-causality.
- **Efficient Market:** Capitalization forces resolve definitively at strictly $t = 0$, fully reflecting option-value instantly upon US News release.

Caveats & Limitations

- **Local Average Treatment Effect (LATE):** The 9.8% premium is intensely localized to the immediate geospatial state border. Housing markets deep in the interior of a state possess considerably lower spatial elasticity due to commuting limits.
- **Prestige Definition:** U.S. News Top 50 is a known, highly publicized public proxy. The genuine structural option value might operate on specific major availability (e.g., highly restricted elite Engineering or Pre-Med programs) embedded solely within those Top 50 schools.

Conclusion & Policy Implications

- **Summary Results:** Spatial equilibrium across US borders heavily prices higher-education prestige, completely neutralizing standard in-state tuition financial savings.
- **Implication for Policymakers:**
  - Broadly subsidizing generic higher education does *not* attract cross-border migration or inflate the local tax base.
  - True geographic brain-drain prevention and profound housing capitalization require sustained, concentrated investment into an elite, globally-recognized flagship capable of breaching the Top 50 threshold.

Appendix A: Pure Price Elasticity Restriction

- Restricted the panel strictly to border segments where Δ **Top 50 = 0**.
- Does the financial price surface when Caliber prestige is identical?

Table: Pure Price Model (Dependent Var: log PPSF)

	(1) Single Cluster	**(2) Two-Way Cluster**
Tuition Gap LATE ($1k)	-0.003	-0.003
	(0.005)	(0.013)
Covariates	✓	✓
Fixed Effects	Seg $\times$ Year	Seg $\times$ Year
Observations	79,318	79,318

Result: A precise, perfectly-identified zero effect for raw financial cost.

Appendix B: Marginal Tax Cliff Restriction

- Restricted panel strictly to dyads located in the lowest income tax differential quartile. Eliminates Tiebout income-tax haven sorting as a confounder.

Table: Tax Restricted Model (Dependent Var: log PPSF)

	(1) Single Cluster	**(2) Two-Way Cluster**
Caliber Gap LATE (Top 50)	0.085*	0.085
	(0.048)	(0.082)
Tuition Gap LATE ($1k)	0.014	0.014
	(0.009)	(0.024)
Covariates	✓	✓
Fixed Effects	Seg × Year	Seg × Year
Observations	36,936	36,936

Appendix C: Stability Across Timeframes

- Does the Remote Work / COVID housing boom rupture our parameters?

Table: Pre-2019 vs Post-2020 Parameters (Two-way Cluster)

	(1) Pre-COVID	**(2) Post-COVID**
Caliber Gap LATE (Top 50)	0.118***	0.062
	(0.040)	(0.041)
Tuition Gap LATE ($1k)	−0.015	−0.001
	(0.015)	(0.010)
Covariates	✓	✓
Fixed Effects	Seg × Year	Seg × Year
Observations	78,046	50,374

Result: Model handles secular shocks well; the Top 50 premium operates strongly across time, though it compresses slightly during national housing spikes.